

Lab Proof: Creative Complaint Handler (LangChain agent)

1. Path chosen

Tool-calling agent, with a guardrail layered on top.

`normalobjects_langchain.py` builds the agent with `create_openai_tools_agent` + `AgentExecutor` (`langchain_classic.agents`), backed by four `@tool`-decorated functions (`consult_demogorgon`, `check_hawkins_records`, `cast_interdimensional_spell`, `gather_party_wisdom`). This is the "agent" path, not plain retrieval, because the LLM decides at runtime which tools to call, in what order, and with what arguments — it isn't a fixed retrieve-then-generate pipeline. A guardrail is bolted onto that agent via the system prompt's "CRITICAL RULE": every factual claim in the final answer must cite a `[SOURCE: ...]` tag, and the model must say plainly when a tool returns `matched_key=NONE` rather than invent a record. That rule is enforced only by prompting, not by code — which is exactly what Section 7/8 test.

2. Environment

- Python 3.14.4 (`.venv/Scripts/python.exe --version`)
- `langchain==1.3.14`
- `langchain-classic==1.0.8`
- Full pinned set in `requirements.txt` (langchain-core 1.5.2, langchain-openai 1.4.1, langgraph 1.2.10, openai 2.50.0, etc.)

3. Setup finding: handout was wrong about imports

Reproduced the failure directly:

```
>>> from langchain.agents import AgentExecutor, create_openai_tools_agent
ImportError: cannot import name 'AgentExecutor' from 'langchain.agents'
(...langchain\agents\__init__.py)
```

Confirmed the fix works:

```
>>> from langchain_classic.agents import AgentExecutor, create_openai_tools_agent
# no error
```

Conclusion: in this pinned version set, `langchain 1.3.14` no longer ships

`AgentExecutor/create_openai_tools_agent` from `langchain.agents` — that surface was split out into the separate `langchain-classic` package. The handout's import path is stale for this LangChain version; the working import is `from langchain_classic.agents import ...`, which is what the script actually uses.

4. The query

My lamp keeps flickering and I think there is a portal in my closet. What do I do?

5. Tool evidence

Real run (AGENT RUN 1, temperature 0.7 — the LLM's tool choices vary between runs, see Section 8):

```
[1] TOOL CALLED: check_hawkins_records
    ARGS: {'query': 'portal'}
    RETURNED: [SOURCE: check_hawkins_records | matched_key='portal' |
query='portal']
HAWKINS LAB FILE 04-B: Portal instability logged Nov 1983. Gate breach originates
in Sublevel 3. Containment: failed. Note: portals do not respond to written
complaints.

[2] TOOL CALLED: check_hawkins_records
    ARGS: {'query': 'electricity'}
    RETURNED: [SOURCE: check_hawkins_records | matched_key='electricity' |
query='electricity']
HAWKINS LAB FILE 02-C: Localized electrical anomalies precede gate activity by 4
to 6 hours. Flickering lights are a leading indicator, not a wiring fault.

[3] TOOL CALLED: consult_demogorgon
    ARGS: {'complaint': 'lamp flickering and possible portal in closet'}
    RETURNED: [SOURCE: consult_demogorgon | input='lamp flickering and possible
portal in closet']
The Demogorgon has filed your complaint in the Upside Down. Expected resolution
time: never, or last Tuesday.

[4] TOOL CALLED: cast_interdimensional_spell
    ARGS: {'problem': 'lamp flickering and possible portal in closet',
'creativity_level': 'high'}
    RETURNED: [SOURCE: cast_interdimensional_spell | problem='lamp flickering and
possible portal in closet' | creativity_level='high' | spells_returned=3]
EGGO BINDING: offer waffles to 'lamp flickering and possible portal in closet'.
Effective on most entities and all children.
CHRISTMAS LIGHT PROTOCOL: spell out 'lamp flickering and possible portal in
closet' on the wall in fairy lights and wait for something to answer.
REVERSAL CHARM: turn 'lamp flickering and possible portal in closet' upside down.
If it was already in the Upside Down, it is now merely inconvenient.
```

6. Final output

It sounds like you have a serious case of "lamp flickering and possible portal in closet," which is not only irregular but potentially dangerous!

Here's what I've found:

1. ****Portals & Flickering Lights****: According to Hawkins Lab records, localized electrical anomalies (like flickering lights) are often a sign of impending gate activity, indicating that your lamp's flicker is not a simple wiring issue, but

rather a leading indicator of a portal nearby. [SOURCE: check_hawkins_records | matched_key='electricity'] Moreover, the records also confirm that any portals in the area have a history of instability and can be quite unpredictable. [SOURCE: check_hawkins_records | matched_key='portal']

2. ****Demogorgon Input****: I consulted the Demogorgon, and it seems your complaint has been filed in the Upside Down. Unfortunately, the expected resolution time is... never, or possibly last Tuesday. So, there's that! [SOURCE: consult_demogorgon | input='lamp flickering and possible portal in closet']

Now, for some creative solutions:

- ****EGGO BINDING****: Offer waffles to the lamp and the closet. This spell is effective on most entities and all young children, so who knows what might happen?
- ****CHRISTMAS LIGHT PROTOCOL****: Spell out "lamp flickering and possible portal in closet" on the wall using fairy lights and wait for a response. You might just get an answer!
- ****REVERSAL CHARM****: Turn the whole situation upside down. If it was already in the Upside Down, it will just be a minor inconvenience instead of a full-blown crisis.

Choose wisely, and good luck!

7. Verification: which claims are grounded

Claim	Source tag cited	Actually supported?
"not only irregular but potentially dangerous"	none	No — editorializing, no tool backs "dangerous"
Flickering lights are a leading indicator of impending gate activity, not a wiring fault	check_hawkins_records matched_key='electricity'	Yes — matches FILE 02-C verbatim
Portals in the area "have a history of instability"	check_hawkins_records matched_key='portal'	Yes — matches FILE 04-B ("Portal instability logged")
Portals "can be quite unpredictable"	same tag as above	Partial — record says instability/failed containment, never says "unpredictable"; a mild extrapolation dressed as a direct citation
Demogorgon filed complaint in the Upside Down, resolution "never, or last Tuesday"	consult_demogorgon	Yes — direct paraphrase of tool return

Claim	Source tag cited	Actually supported?
EGGO BINDING / CHRISTMAS LIGHT PROTOCOL / REVERSAL CHARM spells	<code>cast_interdimensional_spell</code> (implicit, no tag on this claim)	Yes in content, but the final answer never actually attaches a <code>[SOURCE: cast_interdimensional_spell...]</code> tag to these — a guardrail miss even though the content itself is grounded

8. Failure / limitation

The flickering finding: the CRITICAL RULE is enforced by the model choosing to comply, not by any code path that checks it — so both *which* tools get called and *whether* every claim gets tagged flicker between runs of the identical query at temperature 0.7.

Ran the exact same query three times:

- **Run 1** (above): 4 tool calls (`portal`, `electricity`, `demogorgon`, `spell`). Grounded the electricity/flickering claim correctly, but also let one ungrounded claim ("potentially dangerous") and one untagged-but-grounded claim (the spells) through.
- **Run 2**: only 2 tool calls (`portal`, `demogorgon`) — `check_hawkins_records` was never called with `electricity`, so the model had no source for a flickering/wiring claim and correctly avoided making one, but it also never investigated the most literal part of the complaint ("lamp keeps flickering").
- **Run 3**: same 2 tool calls as Run 2 (`portal`, `demogorgon`), same gap.

So the finding reproduced across all 3 runs: tool selection is nondeterministic for an identical input, which means the guardrail's guarantee ("every factual claim comes from a tool result") is only as reliable as the model's per-run choice to (a) call the relevant tool at all and (b) tag every sentence that depends on it. Nothing in the code enforces either.

As a static counterfactual, calling the tool directly on the literal complaint text proves the record exists and would have been available had the agent queried it every time:

```
[SOURCE: check_hawkins_records | matched_key='electricity' | query='my lamp keeps flickering electricity']
HAWKINS LAB FILE 02-C: Localized electrical anomalies precede gate activity by 4 to 6 hours. Flickering lights are a leading indicator, not a wiring fault.
```